

Chp1: Pairwise Sequence Alignment Algorithms

1.1 Edit Graphs, DP, likelihood approx. of scoring schemes

1.2 Global Alignment Alg. 

1.3 Scoring Schemes for Alignment

1.4 Local Alignment Alg.  global alignment of substrings

Algorithm for Local Alignment

 $V(i, j)$: the value of the optimal global alignment between all suffixes of x_i and y_j

$$\text{ALG. } \begin{cases} V(i, 0) = 0 \\ V(0, j) = 0 \\ \text{FOR } i > 0 \text{ AND } j > 0 \text{ DO} \\ \quad V(i, j) = \text{MAX} \begin{bmatrix} V(i-1, j-1) + \delta(x_i, y_j), & \text{if match} = 1 \\ V(i-1, j) + \delta(x_i, -), & \text{if mismatch} = -1 \\ V(i, j-1) + \delta(-, y_j), & \text{INDEL} = -2 \end{bmatrix} \\ V_{\text{OLA}}(x, y) = \text{MAX} [V(i, j), 0 \leq i \leq M, 0 \leq j \leq N] \end{cases}$$

$\rightsquigarrow = 0$

Example: $X = AC$ $\rightarrow$ length: $M=2$
 $Y = GGT$ $\rightarrow$ 2 sequences
 $\rightarrow$ length: $N=3$

 $X_1 = A, X_2 = AC$

$\downarrow$
 prefix
 of size 1
 of X

$\downarrow$
 prefix
 of size 2
 of X

$Y_1 = G, Y_2 = GG, Y_3 = GGT$
 $\downarrow$
 prefixes of size 1, 2, 3 of Y

Alignment of the SUFFIXES of PREFIXES:

 $X_1 = A, Y_1 = G$ suffix $X_1 = A$ suffix $Y_1 = G$ $X_1 = A, Y_2 = GG$ suffix $X_1 = A$ suffix $Y_2 = G, GG$ $X_1 = A, Y_3 = GGT$ suffix $X_1 = A$ suffix $Y_3 = T, GT, GGT$ $X_2 = AC, Y_1 = G$ SUFF: $X_2 = C, AC$ SUFF: $Y_2 = G$ $X_2 = AC, Y_2 = GG$ SUFF: $X_2 = C, AC$ SUFF: G, GG $X_2 = AC, Y_3 = GGT$ SUFF: $X_2 = C, AC$ SUFF: T, GT, GGT $V(1, 1)$: align pairwise all suffixes of X_1 & Y_1 $\rightarrow A$
 G $V(1, 2)$: X_1 & Y_2 $\rightarrow A$ A
 G GG $V(1, 3)$: X_1 & Y_3 $\rightarrow A$ A A
 T GT GGT $V(2, 1)$: X_2 & Y_1 $\rightarrow C$ AC
 G G $V(2, 2)$: X_2 & Y_2 $\rightarrow C$ C AC AC
 G GG G GG $V(2, 3)$: X_2 & Y_3 $\rightarrow C$ C C AC AC AC
 T GT GGT T GT GGT

RECALL: SUFFIXES (PREFIXES (X)) = SUBSTRINGS (X)

 ϵ = empty strings

GLOBAL ALIGNMENT of RANDOM SEQUENCES of size M] $O(M)$
 LOCAL ALIGNMENT of RANDOM SEQUENCES of size N] $O(\log N)$

4 Quantities: we take the max of each of them
 ↳ since one is 0, we don't have negative numbers

CLASS PRACTICE

MATCH = 1

MISMATCH = -1

INDEL = -2

		A	C	T	C	C	A
		0	0	0	0	0	0
A		0	1, -2, -2, 0 1	-1, -2, -2, 0 0	0	0	1, -2, -2, 0 1
C		backtrack 3 to 0 at 3, 0	0	1 ± 1 = 2 2	-1, 0, -2, 0 0	1 - 2 = -2 1	1 - 2, -1, 0 0
A		1 0 1 0 1 1 2 3 = -2 0 1 1 = 1	0, -1, 0, 0 0	0, 0, -1 0	0 - 1 = -1 0, -1, -1 0	1 - 0 = 1 1 - 2 = -1 0 - 2 = -2 0	1 + 1 = 2 -2 2
C		-1, -2, -1, 0 0	1 1 = 2 2, 1 = 2 0	0 - 1 = -1 2 - 1 = 0 1 - 2 = -1 0	1 1 = 2 0 1 = 1 0 1 = 1 2	1 0 = 1 1 - 2 = -1 0 - 2 = -2 1	0 - 1 = -1 2 - 1 = 0 -1 0
T		-1, -2, -2, 0 0	-1, 0, -2, 0 0	2 1 = 3 0 - 2 = -1 0 - 2 = -2 3	0 - 1 = -1 1 - 2 = -1 2 - 2 = 0 1	2 - 1 = 1 1 - 2 = -1 1 - 2 = -1 1	1 - 1 = 0 5 2 = 1 5 2 = 1 0
C		0 - 1 = -1 0 - 2 = -2 0	0 1 = 1 0 2 = 2 0	0 - 1 = -1 1 - 2 = -1 2 - 2 = 0 1	3 1 = 4 1 - 2 = -1 1 - 2 = -1 0	4 1 = 3 4 - 2 = 2 4 - 2 = 2 2	-1 - 2 = -3 2 - 2 = 0 0 - 2 = -2 0

To evaluate the max: vertical, horizontal, diagonal & zero
 Back track from the entry with the max number and stop at the first zero

GAPS & ALIGNMENTS WITH GAPS

DOGMA & MOLECULAR BIO

DNA → RNA → PROTEIN

